1. What is a variable in C/C++?

Ans. A variable is a container which stores a 'value'. Variables in C/C++ stores value of a constant.

2. Why do we use variables in programs?

Ans. To store and manipulate data during program execution.

3. What is the difference between a variable name and its value?

Ans. Variable name: Identifier of the variable.

Value: Data stored in the variable.

4. Which of the following is a valid variable name?

a) 2num

b) total_marks

c) for

d) my marks

Ans. (b) total_marks

5. Why is Age different from age?

Ans. C++ is case-sensitive, so both are different variables.

6. Can a variable name contain spaces? Why?

Ans. No, spaces are not allowed in variable names.

7. Can a variable name start with an underscore (_)?

Ans. Yes, it can.

8. What happens if two variables have the same name in the same scope?

Ans. A compilation error occurs.

9. Write three meaningful variable names for storing student information.

Ans. studentName , studentAge, studentMarks

10. Which naming convention is better and why: studentMarks or sm?

Ans. studentMarks, because it is more meaningful and readable.

11. What is a data type?

Ans. A data type specifies the kind of value a variable can store.

12. Which data type is used to store whole numbers?

Ans. int

13. Which data type is used to store decimal values?

Ans. float or double

14. What is the difference between float and double?

Ans. Double provides greater precision and storage than float.

15. Which data type stores a single character?

Ans. char

16. Which data type stores True/False values?

Ans. bool

17. What data type is suitable for storing a person's name?

Ans. string

18. Predict the output:

```
int age = 18;
cout << age;
```

Output : 18

19. Identify the data type:

25, 3.14, 'A', true.

Ans. 25 → int

3.14 → double/float

'A' → char

true → bool

20. What happens if you store a decimal number in an int variable?

Ans. The decimal part is discarded.

Ex. int x = 9.8;   // so x becomes 9.

21. What is the purpose of cin?

Ans. To take input from the user.

22. What is the purpose of cout?

Ans. To display output on the screen.

23. Which symbol is used with cin?

Ans. >>

24. Which symbol is used with cout?

Ans. <<

25. Explain:

cin >> age;

Ans. Takes input from the user and stores it in age.

26. Explain:

cout << age;

Ans. Displays the value of age.

27. What is the purpose of endl?

Ans. Moves the cursor to the next line.

28. Write a statement to input a student's marks.

Ans. cin >> marks;

29. Write a statement to display "Welcome to C++".

Ans. cout << "Welcome to C++";

30. Predict the output:

cout << "Programming";

Output : Programming

31. What is an operator?

Ans. A symbol that performs operations on operands.

32. Which operator is used for addition?

Ans. +

33. What is the result of 10 % 3?

Ans. 1

34. What is the result of 8 % 2?

Ans. 0

35. Which operator checks equality?

Ans. ==

36. What is the difference between = and == ?

Ans. = is Assignment operator which assigns the value to the variable whereas == is Equality operator which checks equality.

37. What is the result of 5 > 3?

Ans. true (1)

38. What does && represent?

Ans. Logical AND

39. What does || represent?

Ans. Logical OR

40. What does ! represent?

Ans. Logical NOT

41. Write a program to print "Hello World".

Ans.

```
#include <iostream>
using namespace std;
int main()
{
   cout << "Hello World";
   return 0;
}
```

42. Write a program to input two numbers and display their sum.

Ans.

```
#include <iostream>
using namespace std;
int main()
{
   int a, b;
   cin >> a >> b;
```

```
   cout << "Sum = " << a + b;
   return 0;
}
```

43. Write a program to calculate the area of a rectangle.

Ans.

```
#include <iostream>
using namespace std;

int main()
{
   float length, width;
   cin >> length >> width;
   cout << "Area = " << length * width;
   return 0;
}
```

44. What formula is used to calculate Simple Interest?

Ans. Simple_interest = (Principal*Rate*Time)/100

45. Write a program to calculate Simple Interest.

Ans.

```
#include <iostream>
using namespace std;

int main()
{
   float P, R, T, SI;
   cin >> P >> R >> T;

   SI = (P * R * T) / 100;

   cout << "Simple Interest = " << SI;
```

```
    return 0;
}
```

46. Why is a temporary variable used while swapping two numbers?

Ans. To store one value temporarily so it is not lost during swapping.

47. Swap the values:

a = 10, b = 20.

Ans.

```
#include <iostream>
using namespace std;
int main()
{
    int a = 10, b = 20, temp;
    temp = a;
    a = b;
    b = temp;

    cout << "After Swapping:" << endl;
    cout << "a = " << a <<  ", b =  " <<b<<endl;
    return 0;
}
```

48. What formula is used to convert Celsius into Fahrenheit?

Ans. F = (9/5 × °C) + 32

49. What is the Fahrenheit value of 0°C?

Ans.

```
#include <iostream>
using namespace std;

int main()
{
```

```
    int celsius = 0;

    float fahrenheit = (9 / 5.0 * celsius) + 32;

    cout << "Fahrenheit value of 0°C  = " << fahrenheit << " F" << endl;

    return 0;

}
```

50. Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Ans. Because integer division removes the decimal part.

9/5 becomes 1 instead of 1.8.